Ventilation as a bilinear dynamical system

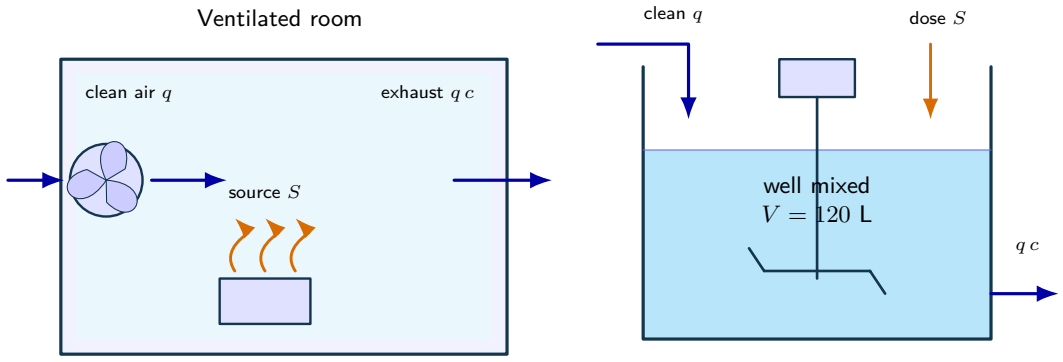
Original learning collection – November 2026

Evidence note. Original teaching scenario. Numerical parameters are deliberately chosen for learning. All generated data and curves are synthetic; no physical measurement is asserted.

1 Design brief

A hypothetical well-mixed room has $V = 120\,\mathrm{m}^3$, tracer generation $S = 360\,\mathrm{ppm\,m}^3/\mathrm{min}$ and nominal airflow $q_0 = 3\,\mathrm{m}^3/\mathrm{min}$. Inlet excess concentration is zero. Derive the equilibrium and local airflow-to-concentration model from $V\dot{c} = S - qc$. Compare 10% and 100% airflow increases using the nonlinear and linearized models, including their final-value errors.

2 Physical system



A ventilated room and a well-mixed liquid analogue share an inventory balance. The room uses m^3 and the liquid case uses L; do not mix their concentration or flow units. Equal inlet and outlet volume flows keep each inventory constant.

3 Concentration balance

For a perfectly mixed room, let c be excess tracer concentration in ppm, q ventilation flow in $\mathrm{m}^3/\mathrm{min}$ and S tracer generation in ppm $\mathrm{m}^3/\mathrm{min}$. Conservation gives

$$V\dot{c} = S - qc.$$

The model is bilinear in the two variables q and c. At constant airflow $q_0 = 3$, the equilibrium is $c_0 = S/q_0 = 120$ ppm.

4 Local perturbation model

Write $c = c_0 + \delta c$ and $q = q_0 + \delta q$. Expanding the product and subtracting the equilibrium gives

$$V\delta\dot{c} = -q_0\delta c - c_0\delta q - \delta q\delta c.$$

Discarding only the second-order product yields

$$\frac{\delta C(s)}{\delta Q(s)} = \frac{-c_0/q_0}{(V/q_0)s + 1} = \frac{-40}{40s + 1}.$$

Increasing airflow reduces concentration, so the negative static gain has a physical meaning.

5 Finite changes and approximation error

For a flow step Δq, the exact response is

$$c(t) = \frac{S}{q_0 + \Delta q} + \left(c_0 - \frac{S}{q_0 + \Delta q} \right) e^{-(q_0+\Delta q)t/V}.$$

The local prediction is $c_{lin}(t) = c_0 - 40\Delta q(1 - e^{-t/40})$. For a 10 percent step the exact final value is 120/1.1 ppm, while the linear prediction is 108 ppm. For a doubling, they are 60 ppm and 0 ppm: a useful warning against extrapolating the tangent model.

6 Interpretation

The linearized model changes neither its time constant nor its input gain after a step. The nonlinear model changes both its equilibrium and decay rate. This is a tracer-transport teaching example, not an indoor-air-quality exposure assessment.

7 What to learn from the experiment

The doubled-flow experiment deliberately leaves the small-signal regime. Its large error is a limitation of linearization, not a numerical failure.

8 System sketch

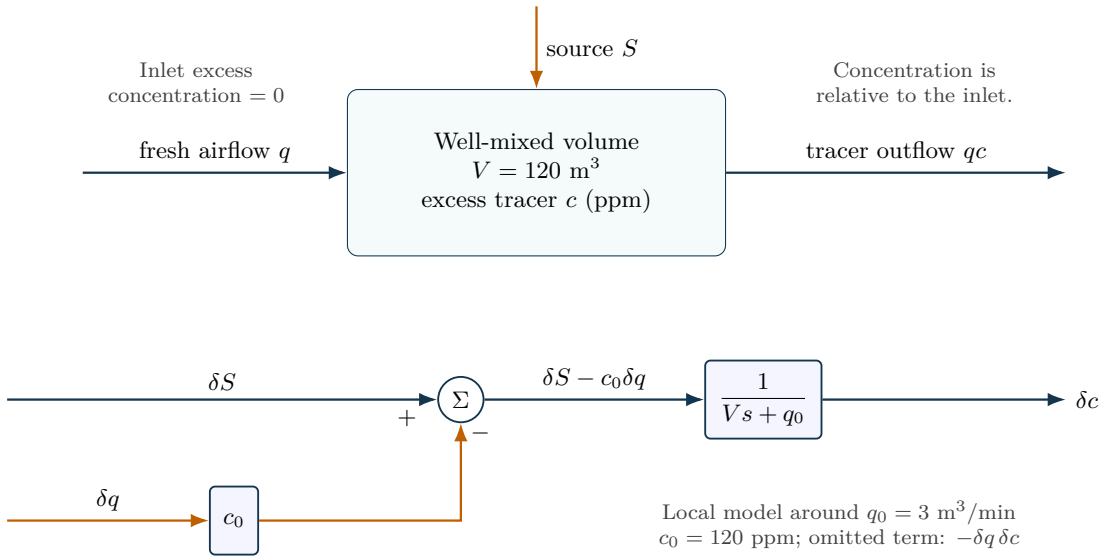
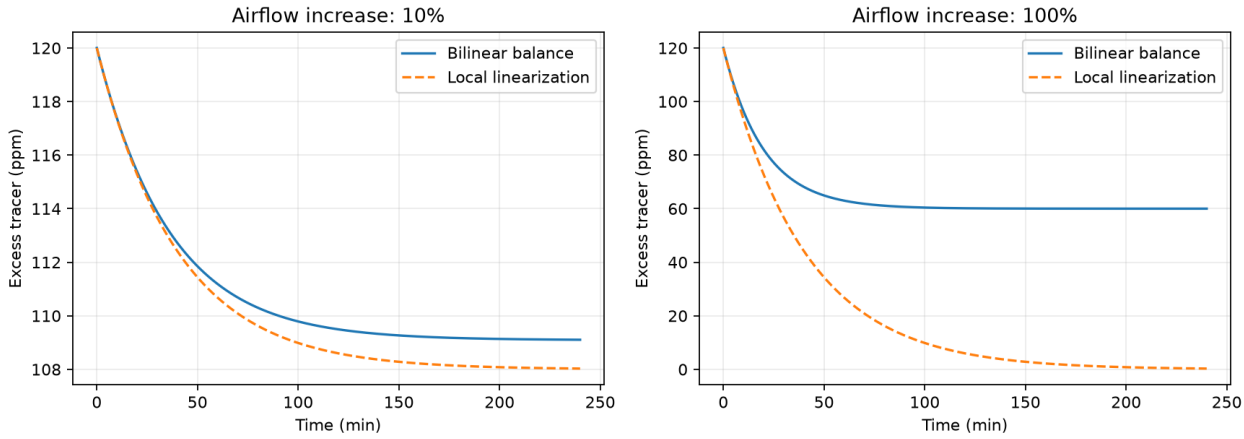


Figure 1. The physical balance and its local two-input signal model. Airflow enters with a negative sign because increased ventilation removes excess tracer.

9 Executed computational evidence

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Airflow +10%: exact final 109.090909 ppm; linear final 108.000000 ppm
Airflow +100%: exact final 60.000000 ppm; linear final 0.000000 ppm
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10 Method references

The equations use standard balance laws and feedback-control methods. Further explanation: https://apmonitor.com/pdc/ and https://fbsbook.org/. Implementation documentation: https://python-control.readthedocs.io/en/0.10.2/. See the repository reference catalog for individual topics and videos.